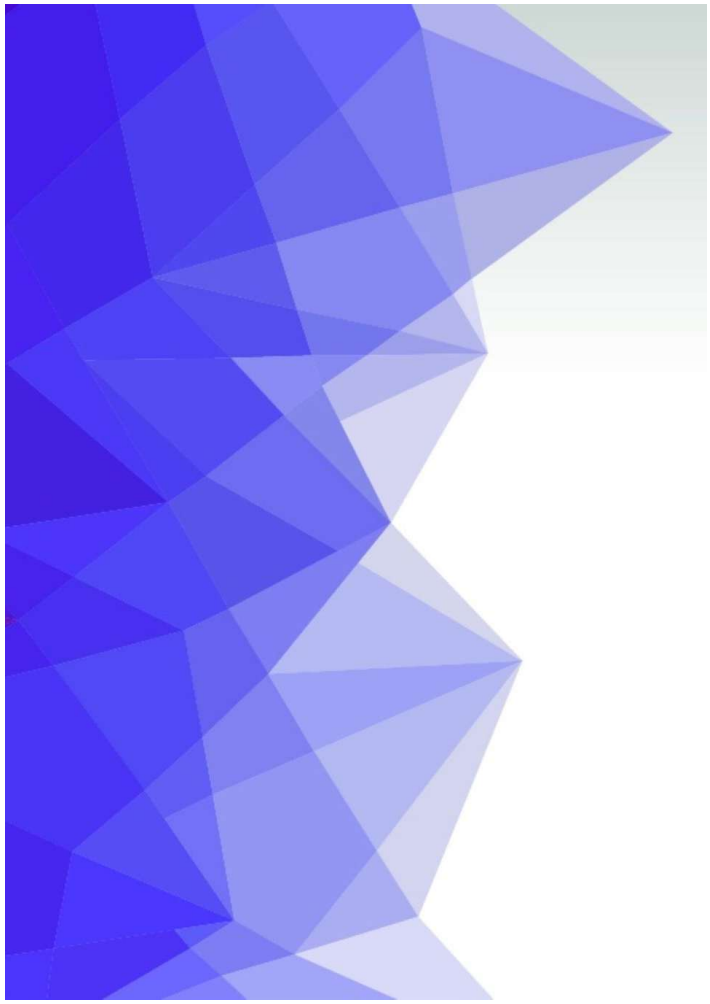


F18CE Week 2

Higher order partial derivatives

Definitions and notations

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = f_{xx} = \lim_{h\to 0}\frac{f_x(x+h,y)-f_x(x,y)}{h}$$
$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} = f_{xy} = \lim_{h\to 0}\frac{f_x(x,y+h)-f_x(x,y)}{h}$$

If f is a function that depends on x and y.

Then f_x, f_y are first order partial derivatives for $f(x,y)$; and

$f_{xx}, f_{yy}, f_{xy}, f_{yx}$ are all second order partial derivatives for $f(x,y)$

Example 2.23

For $f(x,y) = x^3 - 4xy^2$. Find $f_x, f_y, f_{xx}, f_{xy}, f_{yx}, f_{yy}$.

Extra Example (a)

Let $f(x,y) = e^{my}\cos(nx)$, where $m,n \in \mathbb{N}$. Then

$$f_x = -ne^{my}\sin(nx) \qquad f_y = me^{my}\cos(nx)$$

$$f_{xx} = -n^2e^{my}\cos(nx) \qquad f_{yx} = -mne^{my}\sin(nx)$$

$$f_{xy} = -mne^{my}\sin(nx) \qquad f_{yy} = m^2e^{my}\cos(nx)$$

Extra Example (b)

Let $f(x,y) = e^{\alpha x+\beta y}$ where $\alpha, \beta \in \mathbb{R}$ are fixed constant. Then

$$f_x = \alpha e^{\alpha x+\beta y} \qquad f_y = \beta e^{\alpha x+\beta y}$$

$$f_{xx} = \alpha^2 e^{\alpha x+\beta y} \qquad f_{yx} = \alpha\beta e^{\alpha x+\beta y} = f_{xy} \qquad f_{yy} = \beta^2 e^{\alpha x+\beta y}$$

In general, (can prove by induction), for $m, n \geq 0$

$$\frac{\partial^{m+n} f}{\partial x^m \partial y^n} = \alpha^m \beta^n e^{\alpha x+\beta y}\,.$$

Question

If $f(x,y,z) = x^4y^3z^2$. Check that $f_{zyx} = \frac{\partial^3 f}{\partial x \partial y \partial z} = 24x^3y^2z$.

What about $f_{xyz}, f_{xzy}, f_{yxz}, f_{yzx}, f_{zxy}$?

Theorem 2.24 The mixed derivatives theorem

Let $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^2$. If

1. f_x, f_y, f_{xy}, f_{yx} exist at all points in D, and
2. f_{xy} and f_{yx} are continuous on D.

Then $f_{xy}(a,b) = f_{yx}(a,b)$ for all $(a,b) \in D$.

Remarks

1. The mixed derivatives theorem also known as the Clairaut's Theorem or Schwarz's Theorem.
2. We can extend this result to higher order partial derivatives, for example $f_{xyz} = f_{yxz} = f_{zyx} = \cdots$ if all requirements are met.

Example 2.25 (An example when $f_{xy} \neq f_{yx}$)

Consider the following function. We show that $f_{xy}(0,0) \neq f_{yx}(0,0)$ and f_{xy} is not continuous at $(0,0)$.

$$f(x,y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}.$$

To obtain $f_x(0,0)$. We apply the definition

$$f_x(a,b) = \lim_{h\to 0} \frac{f(a+h,b) - f(a,b)}{h}$$

$$f_y(a,b) = \lim_{h\to 0} \frac{f(a,b+h) - f(a,b)}{h}$$

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{f(h,0)}{h} = \lim_{h\to 0} 0 = 0 \quad \because f(x,y) = \frac{xy(x^2-y^2)}{x^2+y^2}$$

$$f_y(0,0) = \lim_{h\to 0} \frac{f(0,h) - f(0,0)}{h} = \lim_{h\to 0} \frac{f(0,h)}{h} = \lim_{h\to 0} 0 = 0 \quad \because f(x,y) = \frac{xy(x^2-y^2)}{x^2+y^2}$$

For $(x,y) \neq (0,0)$, f_x and f_y are given by

$$f_x = y\left[\frac{x^2-y^2}{x^2+y^2} + \frac{4x^2y^2}{(x^2+y^2)^2}\right] \text{ for } (x,y) \neq (0,0). \quad (1)$$

$$f_y = x\left[\frac{x^2-y^2}{x^2+y^2} - \frac{4x^2y^2}{(x^2+y^2)^2}\right] \text{ for } (x,y) \neq (0,0). \quad (2)$$

Compute $f_{xy}(0,0)$ and $f_{yx}(0,0)$ from the definition again:

$$f_{xy}(a,b) = \lim_{h\to 0} \frac{f_x(a,b+h) - f_x(a,b)}{h}$$

$$f_{yx}(a,b) = \lim_{h\to 0} \frac{f_y(a+h,b) - f_y(a,b)}{h}$$

$$f_{xy}(0,0) = \lim_{h\to 0} \frac{f_x(0,h) - f_x(0,0)}{h} = \lim_{h\to 0} \frac{f_x(0,h) - 0}{h} = \lim_{h\to 0} \frac{h(-1) - 0}{h} = -1 \text{ from (1)}$$

$$f_{yx}(0,0) = \lim_{h\to 0} \frac{f_y(h,0) - f_y(0,0)}{h} = \lim_{h\to 0} \frac{f_y(h,0) - 0}{h} = 1 \text{ from (2).}$$

We conclude that $f_{xy}(0,0) \neq f_{yx}(0,0)$.

$$f_{xy} = \frac{(x^2 - 3y^2)(x^2 + y^2) - 2y^2(x^2 - y^2) + 12x^2y^2 - 16x^2y^4}{(x^2 + y^2)^2}$$

Take $x = y$, we have $\lim_{x \to 0} f_{xy}(x, x) = \lim_{x \to 0} \left(-1 + 3 - 4x^2\right) = 2$

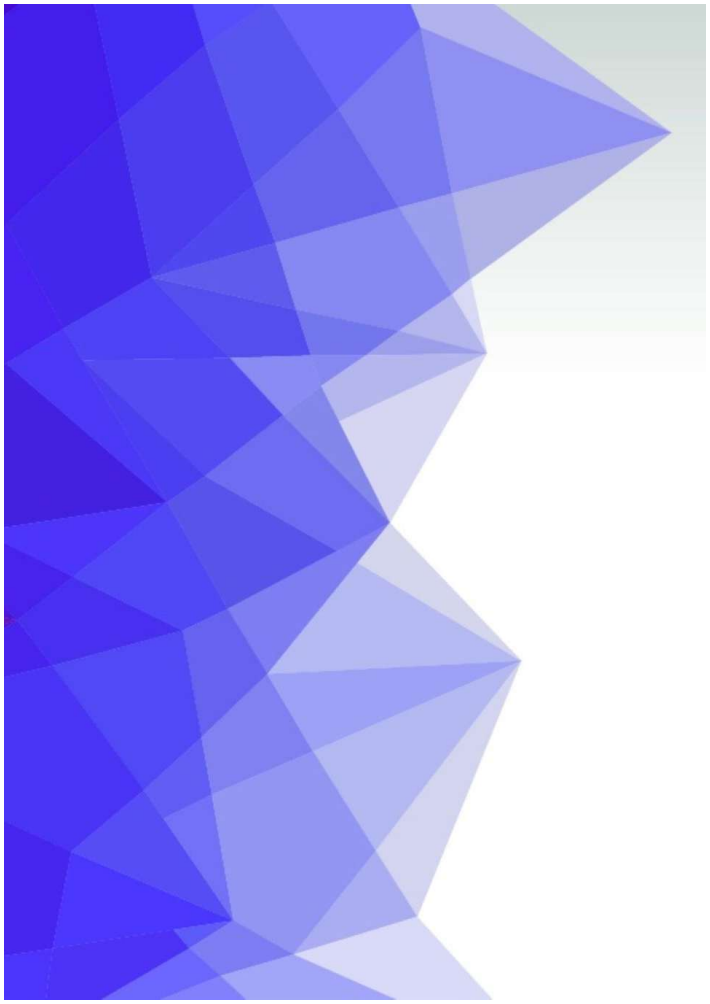
Take $y = x^2$, we have

$$\lim_{x \to 0} f_{xy}(x, x^2) = \lim_{x \to 0} \left(\frac{x^4 + 3x^8 + 8x^6 - 16x^{10}}{(x^2 + x^4)^2} \right) = 1$$

The two answers differ, so limit does not exist and hence the function is not continuous at (0,0).

Factored out the x^4 and simplify to obtain the answer

f_{xy} is not continuous at $(0,0)$.



The chain rules

Theorem 2.29 Chain rule in general

Relationship	Written	Formula	Diagram
f depends on x and x depends on u	$f(x(u))$	$\frac{df}{du} = \frac{df}{dx} \times \frac{dx}{du}$ or $f'(u) = f'(x)x'(u)$ $= f'(x(u))x'(u)$	f | x | u
f depends on x x depends on u, v	$f(x(u,v))$	$\frac{\partial f}{\partial u} = \frac{df}{dx} \times \frac{\partial x}{\partial u}$ or $\frac{\partial f}{\partial v} = \frac{df}{dx} \times \frac{\partial x}{\partial v}$ $f_u = f'x_u,\ f_v = f'x_v$	f | x / \\ u v

f depends on x, y, x depends on u; y depends on u	$f(x(u), y(u))$	$\frac{df}{du} = \frac{\partial f}{\partial x} \times \frac{dx}{du} + \frac{\partial f}{\partial y} \times \frac{dy}{du}$	f / \\ x y | | u u
f depends on x, y, x depends on u, v; y depends on u, v	$f(x(u,v), y(u,v))$	$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \times \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \times \frac{\partial y}{\partial u}$ $\frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \times \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \times \frac{\partial y}{\partial v}$	f / \\ x y / \ / \\ u v u v

f depends on x, y, z each x, y, z depends on u, v, w	$f(x(u,v,w),y(u,v,w),z(u,v,w))$ $\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \times \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \times \frac{\partial y}{\partial u} + \frac{\partial f}{\partial z} \times \frac{\partial z}{\partial u}$ $\frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \times \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \times \frac{\partial y}{\partial v} + \frac{\partial f}{\partial z} \times \frac{\partial z}{\partial v}$ $\frac{\partial f}{\partial w} = \frac{\partial f}{\partial x} \times \frac{\partial x}{\partial w} + \frac{\partial f}{\partial y} \times \frac{\partial y}{\partial w} + \frac{\partial f}{\partial z} \times \frac{\partial z}{\partial w}$	f / | \ x y z / | \ / | \ / | \ uvw uvw uvw

Remarks

1. The variable(s) on the very last row must be all independent one.
2. Distinguish the d (differentiation) and the ∂ (partial differentiation).
3. If f depends on x, then f', f'', f''', etc. all depends on x.
4. If f depends on $x_1, x_2, ..., x_n$, then all partial derivatives (of all orders) depend on $x_1, x_2, ..., x_n$.
5. The idea of the diagram is similar to the tree diagram in probability.

Example 2.26

Let $f : \mathbb{R} \to \mathbb{R}$ and let $u(x, y) = f\left(\frac{y}{x}\right) + 2xy$.

(a) Show that $xu_x + yu_y = 4xy$.

(b) Deduce that $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 4xy$.

Given $u(x,y) = f\left(\frac{y}{x}\right) + 2xy$

(a) Prove that $xu_x + yu_y = 4xy$.

Proof. Let $v = \frac{y}{x}$. Then $v_x = -\frac{y}{x^2}, v_y = \frac{1}{x}$. So,

f
|
v
/ \
x y

$$u_x = \frac{\partial}{\partial x} f\left(v\right) + 2y = f'\left(v\right)v_x + 2y = f'\left(\frac{y}{x}\right)\left(-\frac{y}{x^2}\right) + 2y = -\frac{y}{x^2} f'\left(\frac{y}{x}\right) + 2y$$

$$\Rightarrow xu_x = -\frac{y}{x} f'\left(\frac{y}{x}\right) + 2xy \qquad (1)$$

$$u_y = \frac{\partial}{\partial y} f\left(v\right) + 2x = f'\left(v\right)v_y + 2x = f'\left(\frac{y}{x}\right)\left(\frac{1}{x}\right) + 2x = \frac{1}{x} f'\left(\frac{y}{x}\right) + 2x$$

$$\Rightarrow yu_y = \frac{y}{x} f'\left(\frac{y}{x}\right) + 2xy \qquad (2)$$

$$(1) + (2): \qquad xu_x + yu_y = 4xy$$

b) From (a) we take partial derivative with respect to x and y:

$$\frac{\partial}{\partial x}\left(xu_x + yu_y\right) = \frac{\partial}{\partial x}\left(4xy\right) \Rightarrow u_x + xu_{xx} + yu_{yx} = 4y \Rightarrow xu_x + x^2u_{xx} + xyu_{yx} = 4xy \quad (1)$$

$$\frac{\partial}{\partial y}\left(xu_x + yu_y\right) = \frac{\partial}{\partial y}\left(4xy\right) \Rightarrow xu_{xy} + yu_{yy} + u_y = 4x \Rightarrow xyu_{xy} + y^2u_{yy} + yu_y = 4xy \quad (2)$$

$$(1) + (2): \qquad xu_x + x^2u_{xx} + xyu_{yx} + xyu_{xy} + y^2u_{yy} + yu_y = 8xy$$

Since $xu_x + yu_y = 4xy$, it follows that $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 4xy$.

Example 2.27

Given $f(x,y,z) = \frac{xyz}{(y^2+z^2)^2}$. Find f_{yxzx} away from the points with $y = z = 0$,

i.e., domain of f is $\mathbb{R}^3 \setminus \{(x,0,0) : x \in \mathbb{R}\}$.

Solution. By the mixed derivatives theorem, $f_{yxzx} = f_{xxyz}$ for all $y,z \neq 0$.

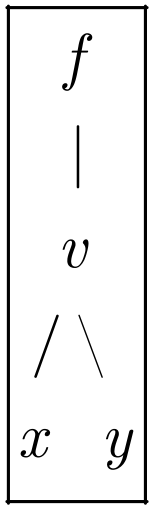
$$f_x = \frac{yz}{(y^2+z^2)^2} \Rightarrow f_{xx} = 0 \Rightarrow f_{yxzx} = f_{xxyz} = 0\,.$$

Why do we choose to start to differentiate partially with respect to x?

Example 2.28

Given $h(x,y)=\Big(f(x-y^2)\Big)^3$. Find h_x, h_y, h_{xy}.

Solution. Let $v=x-y^2$. Then $v_x=1, v_y=-2y$. So,



$$h_x=\frac{\partial}{\partial x}f^3(x-y^2)=3f^2\cdot f'(v)\cdot v_x$$
$$=3f^2\cdot f'(x-y^2)\cdot 1=3\Big(f(x-y^2)\Big)^2\cdot f'(x-y^2)$$
$$h_y=\frac{\partial}{\partial y}f^3(x-y^2)=3f^2\cdot f'(v)\cdot v_y$$
$$=3f^2\cdot f'(x-y^2)\cdot(-2y)=-6y\Big(f(x-y^2)\Big)^2\cdot f'(x-y^2)$$

$$\frac{\partial(f^2)}{\partial y} = 2ff'v_y = -4yff' \qquad \frac{\partial(f')}{\partial y} = f''v_y = -2yf''$$

$$h_{xy} = \frac{\partial}{\partial y}(3f^2f') = 3\left[f'\frac{\partial(f^2)}{\partial y} + f^2\frac{\partial(f')}{\partial y}\right]$$

$$= 3\left[f' \times (-4yff') + f^2 \times (-2yf'')\right] = -6y\left(2f\left(f'\right)^2 + f^2f''\right)$$

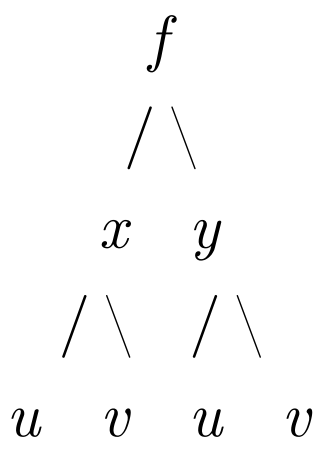
Example 2.30

Let $f(x,y) = x^2 + y,\ x = u^2 + v$ and $y = v - u$.

Find $\frac{\partial f}{\partial u}$ by (i) substitution, and (ii) using the chain rule.

Solution. (i) $f(x,y) = (u^2+v)^2 + v - u$

$$\Rightarrow f_u = 4u(u^2+v) - 1 = 4u^3 + 4uv - 1.$$



(ii) $\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u}$

$= 2x(2u) + 1(-1)$

$= 4ux - 1$

$= 4u(u^2+v) - 1$

$= 4u^3 + 4uv - 1$

Example 2.31

Let f be a function of two variables with continuous first order partial derivatives. A function of three variables h is defined as

$$h(x, y, z) = x(f(x, e^y) + f(xyz, z^2)).$$

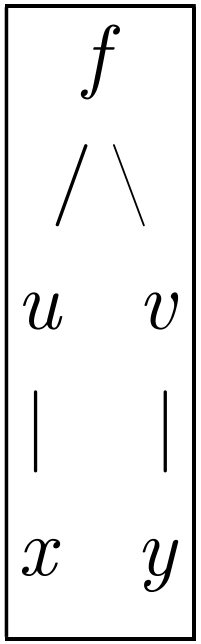
Express the partial derivatives h_x, h_y, h_z in terms of the function f and its partial derivatives. Show all arguments.

Solution.

$$\begin{cases} h_x = f(x,e^y) + f(xyz,z^2) + x\frac{\partial}{\partial x}\Big(f(x,e^y)\Big) + x\frac{\partial}{\partial x}\Big(f(xyz,z^2)\Big) \\ h_y = x\frac{\partial}{\partial y}\Big(f(x,e^y)\Big) + x\frac{\partial}{\partial y}\Big(f(xyz,z^2)\Big) \\ h_z = x\frac{\partial}{\partial z}\Big(f(x,e^y)\Big) + x\frac{\partial}{\partial z}\Big(f(xyz,z^2)\Big) \end{cases}$$

We first find $\frac{\partial}{\partial x}\left(f(x,e^y)\right)$, $\frac{\partial}{\partial y}\left(f(x,e^y)\right)$ and $\frac{\partial}{\partial z}\left(f(x,e^y)\right)$.

Let $f(u,v)=f(x,e^y)$, where $u=x, v=e^y$.

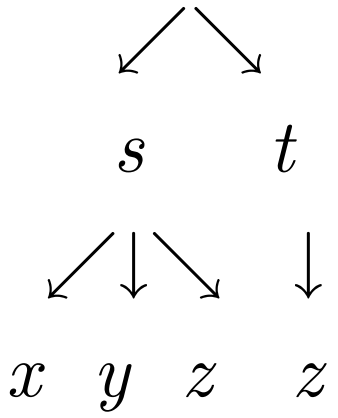


Then $\frac{du}{dx}=1, \frac{dv}{dy}=e^y$.

$$\begin{cases} \frac{\partial}{\partial x}\left(f(x,e^y)\right)=f_u\frac{du}{dx}=f_u(u,v)\times 1=f_x(x,e^y) \\ \frac{\partial}{\partial y}\left(f(x,e^y)\right)=f_v\frac{dv}{dy}=f_v(u,v)\times e^y=e^y f_v(u,v) \\ \frac{\partial}{\partial z}\left(f(x,e^y)\right)=0 \end{cases}$$

Next, we find $\frac{\partial}{\partial x}\left(f(xyz, z^2)\right)$, $\frac{\partial}{\partial y}\left(f(xyz, z^2)\right)$ and $\frac{\partial}{\partial z}\left(f(xyz, z^2)\right)$.

Let $f(s,t) = f(xyz, z^2)$, where $s = xyz, t = z^2$.



Then $\frac{\partial s}{\partial x} = yz, \frac{\partial s}{\partial y} = xz, \frac{\partial s}{\partial z} = xy, \frac{dt}{dz} = 2z$.

$$\begin{cases} \frac{\partial}{\partial x}\left(f(xyz, z^2)\right) = f_s s_x = yzf_s \\ \frac{\partial}{\partial y}\left(f(xyz, z^2)\right) = f_s s_y = xzf_s \\ \frac{\partial}{\partial z}\left(f(xyz, z^2)\right) = f_s s_z + f_t \frac{dt}{dz} = xyf_s + 2zf_t \end{cases}$$

$$\Rightarrow \begin{cases} h_x = f(x, e^y) + f(xyz, z^2) + x\Big(f_x(x, e^y) + yzf_s(xyz, z^2)\Big) \\ h_y = x(e^y f_v(x, e^y) + xzf_s(xyz, z^2)) \\ h_z = x\Big[xyf_s(xyz, z^2) + 2zf_t(xyz, z^2)\Big] \end{cases}$$

where $v = e^y, s = xyz$ and $t = z^2$.

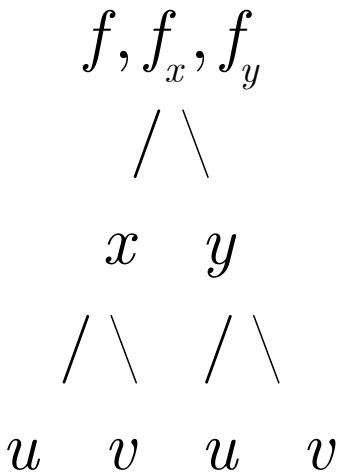
Example 2.32

Let f be a function of x and y with $x = u + v$ and $y = uv$.

(a) Find f_v in term of f_x and f_y.

(b) Given that $f_{xy} = f_{yx}$. Prove that $f_{vu} = f_{xx} + xf_{xy} + f_y + yf_{yy}$.

Solution. Note that f depends on x and y; and so as f_x and f_y.



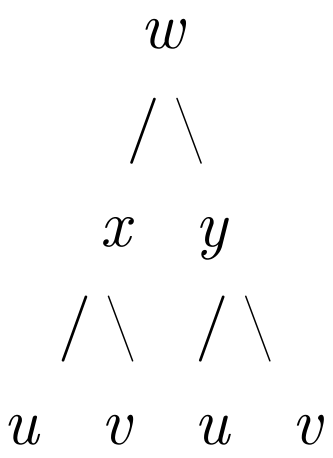
(a) $f_v = f_x x_v + f_y y_v = f_x \times 1 + f_y \times u = f_x + u f_y$

(b) $$\begin{aligned} f_{vu} &= \frac{\partial}{\partial u}\left(f_x + u f_y\right) = f_{xu} + f_y + u f_{yu} \\ &= (f_{xx} x_u + f_{xy} y_u) + f_y + u(f_{yx} x_u + f_{yy} y_u) \\ &= f_{xx} + v f_{xy} + f_y + u f_{yx} + uv f_{yy} \\ &= f_{xx} + (u+v) f_{xy} + f_y + uv f_{yy} \quad (\because f_{xy} = f_{yx}) \\ &= f_{xx} + x f_{xy} + f_y + y f_{yy} \end{aligned}$$

Example 2.33

Let $w(x, y)$ be a function such that $w_x + w_y = 0$.

Suppose $x = u + v$ and $y = u - v$. Prove that $w_u = 0$.



Proof. Given that $w_x + w_y = 0$. Note that $x_u = 1 = y_u$.

Thus, $w_u = w_x x_u + w_y y_u = w_x \times 1 + w_y \times 1 = w_x + w_y = 0$.

Extra example (a)

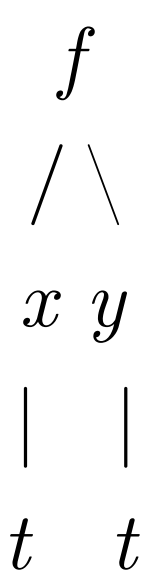
Let $f(x,y) = e^{xy}$, $x(s,t) = s$, $y(s,t) = \cos t$. Find $\frac{\partial f}{\partial s}$.

Solution. $\frac{\partial f}{\partial s} = f_x x_s + f_y y_s = ye^{xy} \cdot 1 + xe^{xy} \cdot 0 = ye^{xy} = \cos t e^{s \cos t}$.

Extra example (b)

Let $f(x,y) = x^2 - y^2$, $x(t) = \cos t$, $y(t) = \sin t$. Find $\frac{\partial f}{\partial t}$.

Solution.
$$\begin{aligned}\frac{\partial f}{\partial t} &= f_x x' + f_y y' \\ &= -2x\sin t - 2y\cos t \\ &= -2\cos t \sin t - 2\sin t \cos t \\ &= -2\sin(2t)\end{aligned}$$



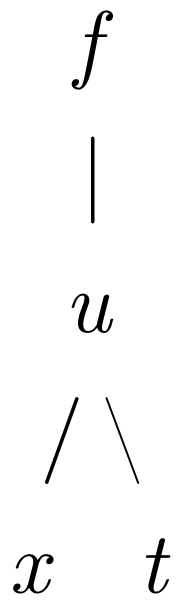
Extra example (c)

Find $\frac{\partial}{\partial t} f(x+ct)$.

Solution. Let $u = x + ct$.

Then $u_t = c$.

$\frac{\partial}{\partial t} f(x+ct) = f' \cdot u_t = cf'(x+ct)$.



Extra example (d)

Prove that $\frac{\partial^n}{\partial t^n} f(x+ct) = c^n f^{(n)}(x+ct), n \in \mathbb{N}$.

Solution. The case of $n = 1$ is proved in the previous slide.

Suppose the result is true for $n = k > 1$, $k \in \mathbb{N}$, i.e.,

$$\frac{\partial^k}{\partial t^k} f(x+ct) = c^k f^{(k)}(x+ct).$$

When $n = k+1$:

$$\begin{aligned}\frac{\partial^{k+1}}{\partial t^{k+1}} f(x+ct) &= \frac{\partial}{\partial t} c^k f^{(k)}(x+ct) \\ &= c^k f^{(k+1)}(x+ct) \times c \\ &= c^{k+1} f^{(k+1)}(x+ct)\end{aligned}$$

By induction, the result is true for all natural numbers n.

